MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4DSCIAM200 – PRINCIPLES OF ROBOTICS AND AUTOMATION

Duration: 1.5 hours Maximum Marks: 35

Practical Examination – Model Question Paper

Instructions:

- **Demonstrate** the given task. (Any one of the tasks listed below)
- **Viva session** (Minimum 5 questions from the Practical Module)
- **Submit Certified Lab Report**

Evaluation Criteria:

- **Viva**: 7 marks
- **Lab Report**: 8 marks
- **Demonstration**: 20 marks

Experiments: (Choose any one)

1. Design a PLC ladder logic program to control a lamp using a push button switch.
2. Develop a PLC program to implement a toggle switch operation.
3. Create a PLC ladder logic program using a timer such that:
    - When a start switch is pressed, an output turns ON after 5 seconds.
    - The output turns OFF immediately when the stop switch is pressed
4. Design a PLC ladder logic program for a latching control circuit.
5. Using PLC ladder logic, simulate the following logic gates:
    - AND gate
    - OR gate
    - NOT gate
6. Design a PLC ladder logic program to control a motor
7. Develop a PLC ladder logic program to simulate a traffic light system
8. Create a PLC program for a temperature control system.

9. Design a PLC ladder logic program for an automatic water level control system.
10. Design a PLC ladder logic program for an alarm system